

RESEARCH ARTICLE

Lattice-Driven Topological Spin Textures in $\text{Cr}_2\text{Ge}_2\text{Te}_6$ Single Crystals

Shuai Dong^{1,2} | Caihong Xie³ | Yuying Bai² | Aile Wang² | Zihao Li² | Xuan Luo⁴ | Yuping Sun^{1,4,5} | Li Pi^{1,2} | Mangyuan Ma⁶ | Yongcheng Deng⁷ | Wenjie Meng¹  | Wei Tong¹ | Yubin Hou¹ | Yalin Lu² | Fu-Hua Sun³ | Qingyou Lu^{1,2,8} | Qiyuan Feng¹

¹Anhui Key Laboratory of Low-energy Quantum Materials and Devices, High Magnetic Field Laboratory, HFIPS, Chinese Academy of Sciences, Hefei, China | ²University of Science and Technology of China, Hefei, China | ³School of Material Science and Engineering, Hubei Key Laboratory of Photoelect Materials and Devices, Hubei Normal University, Huangshi, China | ⁴Key Laboratory of Materials Physics, Institute of Solid State Physics, HFIPS, Chinese Academy of Sciences, Hefei, China | ⁵Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing, China | ⁶State Key Laboratory of Low-Dimensional Quantum Physics and Department of Physics, Tsinghua University, Beijing, China | ⁷State Key Laboratory of Semiconductor Physics and Chip Technologies, Institute of Semiconductors, Chinese Academy of Sciences, Beijing, China | ⁸Hefei National Laboratory, Hefei, China

Correspondence: Qingyou Lu (qxl@ustc.edu.cn) | Qiyuan Feng (fqyuan@hmfl.ac.cn)

Received: 25 November 2025 | **Revised:** 16 January 2026 | **Accepted:** 20 January 2026

Keywords: $\text{Cr}_2\text{Ge}_2\text{Te}_6$ | magnetic anisotropy | magnetic force microscopy | spin waves | topological spin textures | van der Waals magnet

ABSTRACT

Owing to their unique crystal structures, van der Waals (vdW) materials provide a versatile platform for exploring emergent physical phenomena and enabling next-generation electronic and spintronic applications. However, the direct correlations between vdW crystal and intrinsic properties, such as magnetism, remain largely unexplored. Here, we demonstrate that a variety of structure-correlated topological spin textures, characterized by highly regular shapes ranging from triangles to octagons and arranged in periodic patterns, can be induced in vdW crystals. Owing to their well-resolved boundaries, the entire evolution of the quasiparticle characteristics of the topological structures, including creation, structural distortions, and collision were unambiguously revealed. More importantly, it was found that the topological annihilation occurs in an explosive manner, providing direct confirmation of the quasiparticle nature at the scale of a single magnetic unit. Simulations reveal that their formation arises from the interplay between uniaxial anisotropy and dipolar interactions, further modulated by the lattice background. These findings uncover a previously unrecognized structure-property relationship in vdW magnets and open avenues for designing and tuning of new topological spin textures.

1 | Introduction

The interaction between lattice and electronic degrees of freedom plays the most important role in solids and is particularly pronounced in van der Waals (vdW) materials. Since their discovery, the distinctive weakly coupled layered structures of vdW crystals have been closely linked to a variety of emergent

electronic phenomena, highlighting the importance of lattice-electron coupling in these systems. A seminal example is the electric field effect in graphene, where the unique 2D honeycomb lattice of carbon atoms directly modifies the electronic band structure, allowing gate-tunable carrier densities and giving rise to unconventional quantum transport phenomena [1–4]. Along this direction, intensive research has subsequently focused on

Shuai Dong and Caihong Xie have contributed equally to this work.

© 2026 Wiley-VCH GmbH

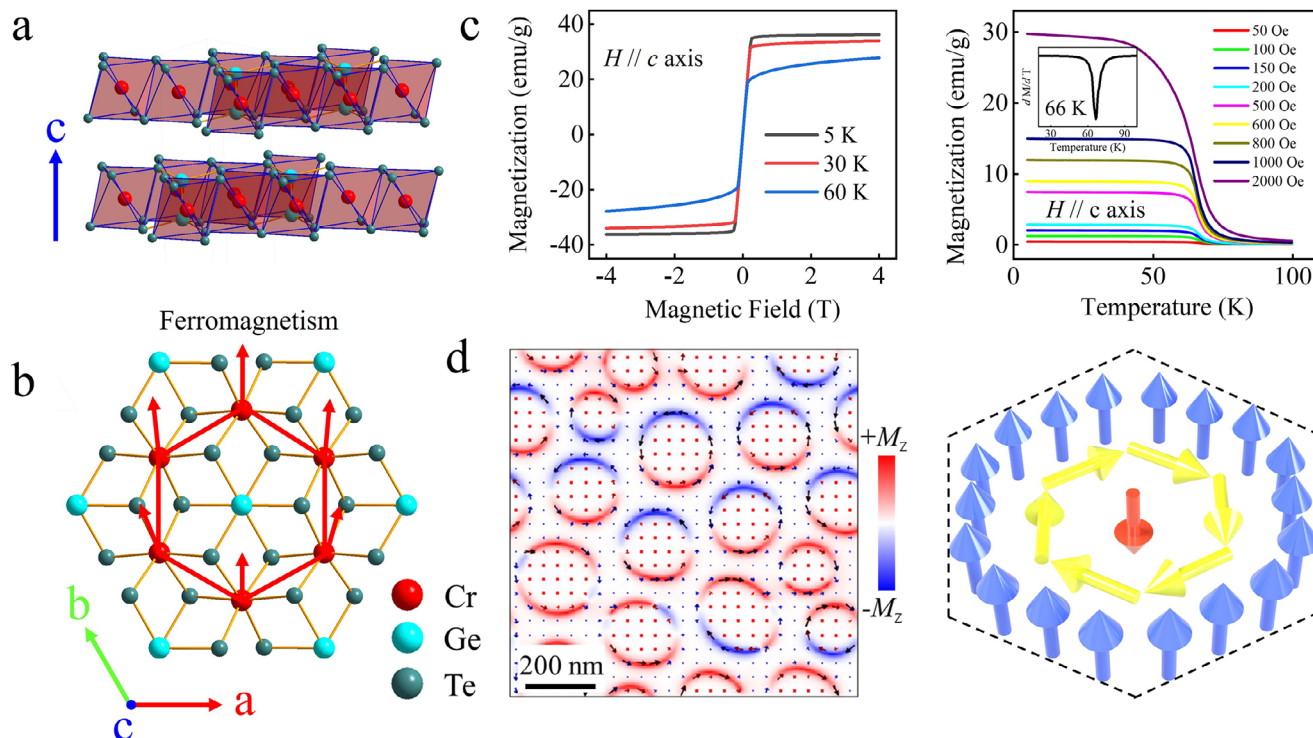


FIGURE 1 | Crystal structure and magnetic properties of CGT. (a) View perpendicular to the c axis, showing the CrTe_6 octahedral structure. (b) View along the c axis, illustrating the ferromagnetic honeycomb network formed by Cr atoms. The unit is outlined by a solid red hexagon. (c) Left panel: Magnetization as a function of magnetic field at different temperatures. Right panel: Temperature dependence of the magnetization under fields ranging from 50 to 2000 Oe. The inset shows the dM/dT curve under 100 Oe FC. The magnetic field is applied along the c axis in all measurements. (d) Left panel: Hexagonally arranged skyrmionic bubbles in CGT, obtained from micromagnetic simulations. Right panel: Hypothetical lattice-correlated topological spin textures.

manually tuning or designing 2D lattice structures, such as engineering moiré superlattices or stacking multilayer structures in specific configurations. This approach enables precise control over the electronic band structure, such as the induction of flat bands, thereby giving rise to a range of exotic quantum phenomena, including the fractional quantum anomalous Hall effect and unconventional chiral superconductivity [5–8].

Beyond electron-charge degree of freedom, the lattice can also strongly couple with spin, leading to a rich array of spin-dependent phenomena. 2D magnetic systems can inherit the magnetocrystalline anisotropy arising from lattice-spin interactions in their bulk counterparts. This anisotropy plays a crucial role in suppressing thermal fluctuations that would otherwise disrupt long-range magnetic order, as predicted by the Mermin-Wagner theorem, thereby enabling the persistence of magnetism down to the monolayer limit [9–12]. Again, analogous to the tuning of the electronic charge degree of freedom, modulating the periodicity of the vdW lattice structure offers a powerful route to reconstruct magnetic properties, giving rise to emergent phenomena like moiré magnetism and novel spin textures [13–16]. Despite significant advances in imaging and controlling these textures, a critical question remains largely unexplored: How does the intrinsic crystallographic structure of vdW magnets influence the formation, morphology, and dynamics of spin textures [13, 17–19]? While much effort has been devoted to engineering spin textures via external stimuli such as electric currents, magnetic fields, or strain, the role of lattice symme-

try and structural environment has yet to be fully elucidated experimentally [17, 20–29].

Owing to the weak interlayer coupling and pronounced lattice anisotropy inherent in vdW magnets, structure-correlated spin textures may readily form even in high-quality crystal flakes [13, 17, 18, 22–30]. In this study, $\text{Cr}_2\text{Ge}_2\text{Te}_6$ (CGT) bulk single crystals were selected for magnetic imaging in the view of their relatively strong ferromagnetic signals. More importantly, the emergence of diverse and nontrivial spin textures in CGT is strongly correlated with the underlying lattice structure, owing to pronounced anisotropic spin-lattice coupling [9, 18, 23, 31–35]. Using real-space imaging, we show that ordered mesh-like spin textures with topologically protected properties can be induced in a vdW magnet through the strengthening of magnetic anisotropy. This is realized either by lowering the temperature to enhance the intrinsic magnetocrystalline anisotropy, or by applying a small external magnetic field along the easy axis to serve as an effective anisotropy [36]. As a result, the observed spin textures exhibit a regular hexagonal unit shape and further organize into an edge-sharing hexagonal superlattice, which is strongly correlated with the intrinsic lattice symmetry of CGT. The influence of domain wall energy and spatial confinement enables the hexagonal spin textures (HSTs) to adopt various polygonal configurations, ranging from triangles to octagons, all exhibiting the same topological stability. The dynamic formation and collapse of the topological spin textures at the single-object level have also been fully captured [28].

CGT crystallizes in a layered vdW structure with rhombohedral symmetry and belongs to the space group $R\bar{3}$ in the hexagonal setting [37]. As shown in Figures 1a,b, within each layer, the Cr atoms form a 2D honeycomb lattice composed of edge-sharing CrTe_6 octahedra. Although a weak direct antiferromagnetic exchange exists between in-plane neighboring Cr atoms, the dominant Cr–Te–Cr superexchange interaction favors ferromagnetism due to the nearly 90° bonding geometry (Goodenough-Kanamori rules). The Cr–Cr atoms define a well-ordered network of hexagonal rings, which are essential to the magnetic interactions in CGT (Figure 1b) [38–40]. In low temperatures, the intrinsic magnetocrystalline anisotropy generated by spin-orbit coupling (SOC), together with the distortion of the CrTe_6 octahedra along the c axis, facilitates an out-of-plane magnetic easy axis and stabilizes ordered ferromagnetism [41]. This unique magnetic anisotropy mechanism renders CGT highly susceptible to lattice-coupled anisotropic stimuli such as electric current, pressure, and magnetic fields, even in its bulk form [18, 41–43].

As shown in the left panel of Figure 1c, CGT is a soft ferromagnet with negligible coercivity. Note that the out-of-plane saturation magnetization increases monotonously with decreasing temperature, which reflects enhanced spin alignment along the c -axis. According to the Callen-Callen power law, the uniaxial anisotropy constant scales with the saturation magnetization, indicating a concomitant increase in magnetic anisotropy at lower temperatures. This is further supported by broadband ferromagnetic resonance (FMR) measurements, which show that the uniaxial magnetocrystalline anisotropy constant of bulk CGT increases fivefold as the temperature decreases from 60 K to 2 K (0.77×10^5 erg/cm³ at 60 K to 3.95×10^5 erg/cm³ at 2 K) [44, 45]. The enhanced magnetic anisotropy in CGT is promoted by increased lattice distortion of the CrTe_6 octahedra, in conjunction with the strong spin-orbit coupling arising from Te atoms. These effects cooperatively favor out-of-plane spin alignment and contribute to the stabilization of the ferromagnetic ground state. Apart from the temperature dependence, the external magnetic field can act as an effective anisotropy field, favoring out-of-plane spin orientation and thus enhancing the magnetization along the c -axis, as evidenced in the right panel of Figure 1c.

Magnetic anisotropy plays an essential role in both 2D and bulk regimes. In 2D systems, it overcomes thermal fluctuations and stabilizes ferromagnetism down to a few atomic layers [9]. Moreover, adjusting the magnetic anisotropy can effectively influence critical parameters of 2D magnetism, including the Curie temperature (T_C), and can directly induce transitions between different spin configurations [9, 43, 46–48]. For instance, the recently developed cantilever-based superconducting quantum interference device (SQUID) scanning technique offers a powerful approach that combines ultrahigh magnetic field sensitivity with high spatial resolution [49]. Utilizing the SQUID-on-lever magnetic imaging technique, the thickness-dependent magnetic ordering in CGT was successfully resolved down to the few-layer limit [12]. Moreover, complex magnetic configurations, such as skyrmionic bubbles, were also observed in few-layer CGT.

The results reveal that the magnetic texture of CGT is strongly influenced by the competition between crystalline and shape anisotropies.

In bulk systems, it has been widely reported that hexagonally packed topological spin textures, such as magnetic skyrmions, can be stabilized by magnetic anisotropy, as demonstrated by the micromagnetic simulation results shown in the left panel of Figure 1d [17, 20, 23, 24, 27, 28, 50, 51]. More detailed information about the micromagnetic simulations can be found in Figures S1 and S2 of the Supporting Information. Such a skyrmion lattice is also observed in our magnetic force microscope (MFM) measurements, as shown in Figure S3 of the Supporting Information. However, it has been widely reported that in layered materials with tetragonal, hexagonal, or trigonal crystal structures, such as CGT, it is highly likely that topological spin textures inherently carry a strong lattice imprint and can further organize into ordered lattice-like configurations (right panel of Figure 1d) [30, 52, 53].

3 | Temperature-Driven Hexagonal Topological Spin Textures in CGT

To directly reveal the lattice-coupled topological spin textures in CGT, a home-built low-temperature MFM was employed to track the evolution of spin textures during the zero-field cooling (ZFC) process. Our home-built MFM enabled high-resolution imaging of a variety of delicate topological magnetic structures and was also well suited for the present study [54–58]. More details about the ZFC process can be found in the Figure S4. As shown in Figure 2a, no magnetic contrast was observed above 75 K, and the entire image shows a homogeneous background, consistent with a paramagnetic ground state. At 71.4 K, cloud-like patterns emerged, but no discernible domain structures were observed. This can be attributed to the short-range ferromagnetic correlations established above the T_C , which have been widely reported by electron spin resonance (ESR) and heat capacity measurements, or inferred from Curie-Weiss fitting of the susceptibility, yet have never been directly visualized [43, 45, 59]. However, with a further decrease of temperature of only 0.2 K, long-range periodic stripe domains with weak magnetic contrast first appeared in the bottom-right corner in Figure 2c, and further expanded to occupy the entire sample area with an additional 0.2 K decrease (Figure 2d). The onset temperature of ordered ferromagnetism observed by MFM (71.2 K) is slightly higher than that obtained from magnetization measurements (66 K), as shown in the inset of Figure 1c. The slightly higher onset temperature observed by MFM may arise from its sensitivity to local magnetic ordering, which can emerge prior to the establishment of ferromagnetism detectable by bulk magnetization measurements. More importantly, the abrupt magnetic transition within an extremely narrow 0.4 K range highlights the strong magnetoelastic coupling in CGT, where even minimal lattice distortions in the CrTe_6 octahedra markedly influence magnetic anisotropy and ordering [43].

With further lowering of the temperature to 69.5 K, an elliptical closed-loop magnetic structure forms, as indicated by the red dashed rectangular box. It further deformed into a distorted

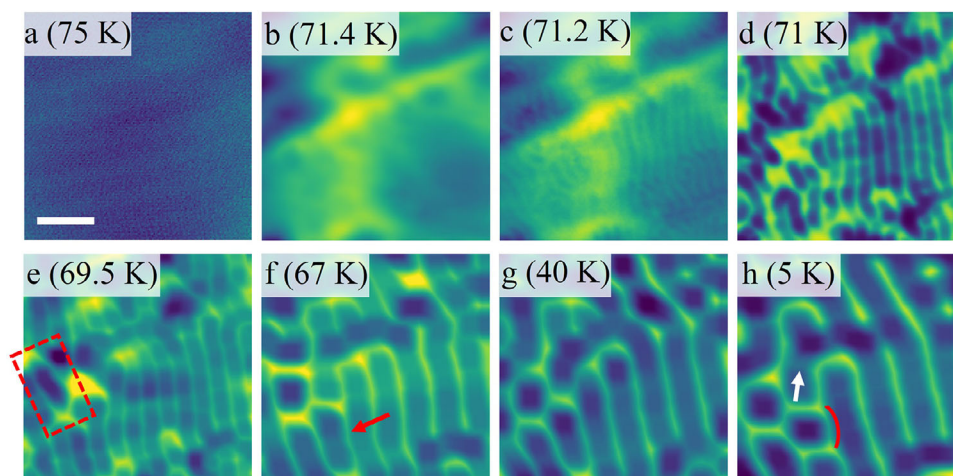


FIGURE 2 | MFM images acquired during the ZFC process. All images were taken at the same locations with the same scan scale of 8 μm . (a) MFM image taken well above T_C , showing a paramagnetic background. (b–h) Temperature-dependent MFM images demonstrate the changes in the intrinsic ferromagnetic ground state. The scale bar is 2 μm . The length of the edge of the hexagonal ring in Figure 2h is about 900 nm.

pentagonal structure at 67 K and finally transformed into a regular hexagonal structure below 40 K. It should also be noted that an elongated closed-loop magnetic structure develops adjacent to the aforementioned hexagonal structure at 67 K, as marked by the red arrow. As the temperature decreases, the structure gradually shrinks along the elongated edge and eventually transforms into a distorted pentagonal shape at 5 K. Based on the MFM observations and insights from the energy landscape, we propose its microscopic formation mechanism. Closer inspection of the pentagonal structure's boundary reveals that its right edge is pinned by a continuous, elongated strip domain. As indicated by the red solid arc in Figure 2h, although this edge already bends outward and tends to deform into a hexagonal structure, the magnetocrystalline anisotropy is not sufficient to break this strip domain, preventing the formation of a regular hexagonal ring. In addition to the continuously transformation from a ring-shaped structure into a hexagonal configuration, the strip domain can also evolve into a hexagon through continuous deformation. As indicated by the white arrow in Figure 2h, an incomplete hexagon has already formed and most of its edges have already arranged in an ordered form. At the missing boundary, a subtle edge is discernible, suggesting that spins in this region are beginning to assemble into an edge under the influence of magnetic anisotropy. Based on these observations, it can be naturally inferred that an edge-sharing hexagonal lattice may form with the further enhancement of lattice effects.

To prove this point, we subsequently performed field-cooling (FC) measurements under various magnetic fields, and the results are summarized in Figure 3a. In this work, all magnetic fields are applied along the magnetocrystalline anisotropy direction, namely the c axis. Since little change occurs below 60 K, we directly present the FC results at 60 K. The starting temperatures of the FC process is 120 K, which is well above the T_C , ensuring that any remanent magnetic moments from the previous thermal cycle are fully eliminated. With the application of a small field of 50 Oe, the strip domains that dominate in the ZFC process disappear, and the entire sample becomes occupied by polygonal and elongated polygonal structures. Unclosed ring-like structures

still exist. However, at the broken edges, magnetic imaging clearly reveals that the spins are assembling to form a complete boundary, as indicated by the red arrow. In addition, aside from the hexagonal structures, 3-, 4-, and 5-membered ring structures are abundant and remain dominant, as marked by the black numbers within the rings. The proportion of hexagonal rings is relatively low, accounting for only 18.8%. Dramatic changes were observed during the FC process upon the application of a 100 Oe magnetic field, leading to the formation of an edge-sharing polygonal lattice structure. This magnetic pattern is very similar to the hexagonal moiré magnetism in twisted double-trilayer CrI_3 [13]. It has been reported that the transition temperature of CGT atomic layers is highly sensitive to the magnetic field (around 0.3 T). The applied out-of-plane field can act as an effective anisotropy, countering thermal fluctuations and enhancing T_C in the 2D regime [9]. Although the bulk material exhibits negligible dependence of T_C on magnetic field (a field exceeding 10 T is required), our results indicate that even a small field of 100 Oe can enhance the anisotropy in the bulk form, thereby altering the magnetic ground state and promoting the formation of lattice-correlated spin structures.

The statistics of the percentage of HSTs are shown at the top of each image (see Table S1). Upon further increase of the magnetic field, the hexagonal rings become increasingly well-defined, and the fraction of occupied hexagons steadily rises. This is further supported by the fast Fourier transform (FFT) of each panel displayed in the lower right corner. At 50 Oe, the FFT pattern displays an elongated hexagonal geometry, likely arising from the elongated polygonal magnetic structures. As the field is increased to 100 Oe, the FFT pattern evolves into a circular shape, which can be attributed to the mixed state of polygonal structures, where the occupancy of hexagonal rings is less than 50%. Between 100 and 600 Oe, the FFT patterns evolve into a highly regular hexagonal lattice with six-fold symmetry, particularly prominent at 200 Oe and 500 Oe. At 800 Oe or 1000 Oe, the images are quite similar to that at 50 Oe. Elongated polygonal structures appear again and this should be ascribed to the emerge of adject hexagonal rings by the relatively large magnetic field. To quantitatively analyze the intrinsic periodicity and symmetry, the 2D autocorrelation map of

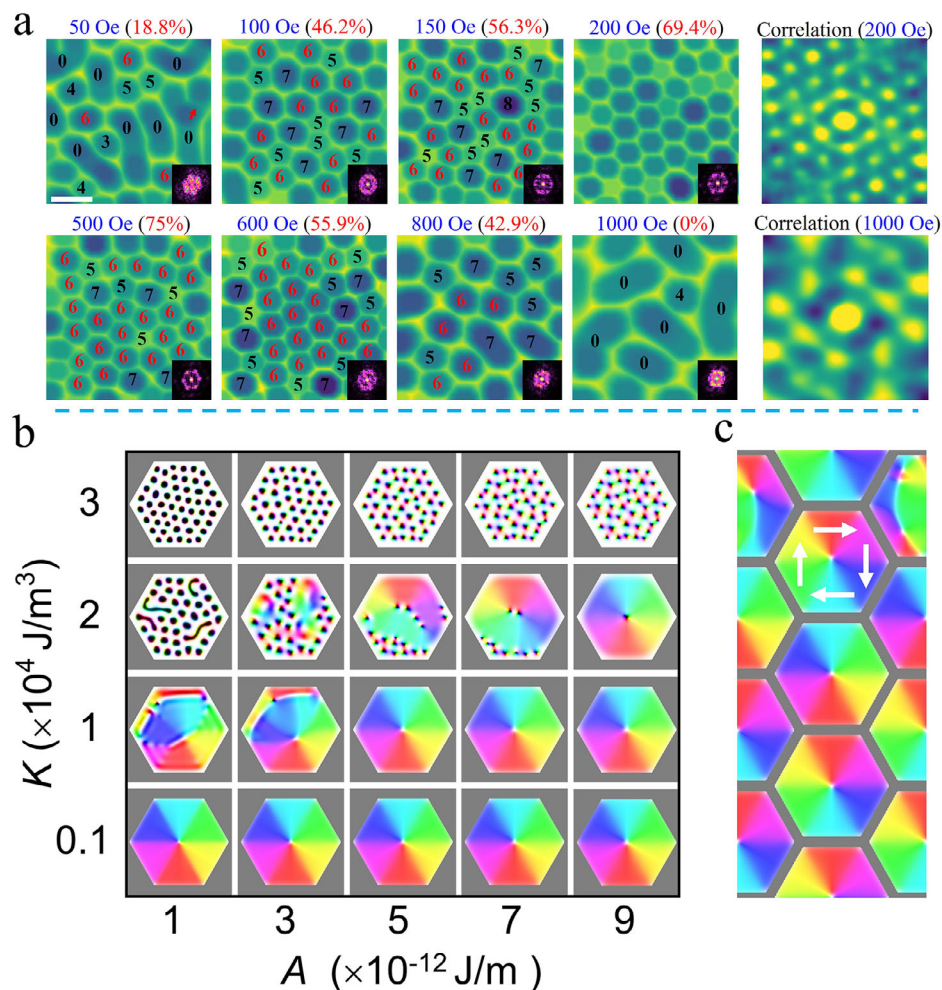


FIGURE 3 | (a) MFM images at 60 K after field cooling from 120 K at various applied fields. The title at the top of each image is formatted as: field (percentage of HSTs). Each hexagonal ring is labeled with a red number 6 positioned at its center. All non-hexagonal rings are labeled with black numbers indicating their number of edges. Note that the elongated polygonal structures are classified as circular spin structures and labeled with the number 0. To ensure accuracy, rings at the periphery of the images with ambiguous or indiscernible edge numbers are excluded from consideration. The FFT pattern corresponding to each image is shown in the lower right corner. The scan scale is 8 μm and the scale bar is 2 μm . The characteristic length scale of the edge of the hexagonal ring is about 800 nm. The 2D autocorrelation maps at 200 Oe (top) and 1000 Oe (bottom) are shown. The lattice constant is approximately 800 nm, which agrees well with the real-space observations. (b) and (c) Micromagnetic simulations of the HST and HST lattice, respectively. The side length of the hexagon is 600 nm, which is comparable to the experimental measurement. The white arrows indicate the direction of the in-plane magnetization.

the magnetization is shown. At 200 Oe, the map displays a well-ordered hexagonal lattice with a lattice constant of approximately 800 nm, which is comparable to the edge length of the hexagonal rings observed in the MFM images. By contrast, the map corresponding to 1000 Oe exhibits a significantly distorted hexagonal arrangement.

In both the ZFC and FC processes, however, the formation of the HSTs should be dominated by the lattice-driven mechanism [55, 60]. Although the images in Figure 3a were obtained across different temperature cycles, they were acquired at exactly the same sample location by matching identifiable surface topographic features (see supplementary Figure S5). The crystal used for imaging was mechanically cleaved to obtain a pristine, atomically flat surface. Therefore, the effects arising from spatial variation or crystal defects can be fully excluded. Furthermore, the formation of hexagonal

textures during ZFC process, as well as the increase in their occupation ratio under larger magnetic fields during FC, are both directly correlated with the enhancement of magnetic anisotropy. Note that magnetic anisotropy actually composed by magnetocrystalline anisotropy and magnetic dipolar anisotropy [23, 60]. Magnetic dipolar anisotropy is highly shape-dependent and generally favors in-plane magnetization in thin flake samples, such as the quasi-2D CGT vdW magnet. In our study, we mainly focused on the control of magnetic textures in bulk CGT through temperature- and magnetic-field-induced modulation of magnetic anisotropy. The work by Vervelaki et al. directly demonstrates that the layer-dependent magnetic textures are also governed by the thickness-dependent balance between crystalline and shape anisotropies [12]. From this perspective, the formation of out-of-plane magnetized hexagonal textures should be attributed to the intrinsic magnetocrystalline anisotropy. A stronger magnetocrystalline

anisotropy implies that the spin orientations are more tightly constrained by the crystalline potential wells, leading to enhanced spin-lattice coupling. Consequently, the symmetry of the underlying lattice facilitates the emergence of hexagonal magnetic textures [30].

This viewpoint is further supported by the micromagnetic simulations shown in Figures 3b,c, in which the symmetry effect is mimicked by the nanoscale geometric confinement. Note that boundary confinement on the scale of hundreds of nanometers is insufficient to model an atomic lattice whose periodicity is only a few angstroms. In our manuscript, we emphasize that lattice symmetry plays a crucial role in the formation of the hexagonal skyrmion lattice. The micromagnetic model with mesoscale hexagonal confinement is therefore employed to capture the essential symmetry-driven energetics at larger length scales. Although the atomic lattice periodicity is far smaller than the magnetic texture, numerous studies have shown that the crystal lattice can substantially influence the symmetry and stability of skyrmion lattices and related spin textures in real materials. For example, theoretical work on frustrated hexagonal magnets demonstrates that the interplay between lattice symmetry and magnetic interactions stabilizes 3Q phases and skyrmion crystal states with hexagonal symmetry in the phase diagram, highlighting the crucial role of underlying lattice symmetry in determining magnetic superstructures [61]. Therefore, although our micromagnetic model does not explicitly resolve atomic-scale periodicity, it effectively incorporates the influence of crystal symmetry originating from the atomic lattice, which is the relevant factor governing the emergence of large-scale hexagonal magnetic patterns.

Notably, a substantial number of non-hexagonal polygons, ranging from triangles to octagons, are also present. Their formation could result from the following two factors. The basic original reason should be the competition between the magnetocrystalline anisotropy energy and the magnetic domain wall energy [62]. Due to the pinning effect induced by magnetic defects or the confinement imposed by stripe domains, polygonal structures with fewer than six edges can form, as discussed in the ZFC process. For example, a pentagonal texture appears during the ZFC process in Figure 2, and a quadrilateral is observed under the 50 Oe field-cooling condition in Figure 3a. The presence of quadrilateral or pentagonal units can further disrupt the regularity of the hexagonal mesh, and under lateral geometric confinement, leads to the emergence of larger polygonal rings such as heptagons and octagons. For instance, under a 150 Oe FC condition (Figure 3a), an octagonal ring is formed, surrounded by eight adjacent polygons. Similar results, the heptagonal structures are formed under 200 Oe, 500 Oe, and 600 Oe FC conditions, as shown in Figure 3a. Notably, under the 50 Oe FC condition, heptagons and octagons cannot form due to the absence of the mesh lattice, which further supports our interpretation. Although they exhibit different morphologies, these structures are intrinsically governed by magnetocrystalline anisotropy and are topologically equivalent in nature. These observations reflect the intricate interplay between topological constraints and geometric frustration within the spin texture network.

4 | Field-Enhanced Formation of Hexagonal Topological Spin Texture Lattices

Recognizing that the applied field can act as an effective magnetic anisotropy, we expect that the formation of HSTs may also be primarily driven by the field. To investigate this scenario, two field-cycling measurements starting from different initial magnetic states were conducted at a relatively high temperature of 60 K to minimize the influence of lattice effects. In addition, this allows for a direct comparison with the series of FC measurements. We first used the 50 Oe field-cooled image shown in Figure 3a as the initial state and subsequently performed in situ MFM measurements while sweeping the magnetic field up to 1000 Oe. As shown in Figure S6, the elongated strip-like polygonal structures become increasingly segmented by emerging boundaries and eventually evolve into a regular lattice structure with increasing magnetic field, similar to the results observed during the FC process at higher fields. This indicates that the magnetic field can indeed enhance the magnetic anisotropy and amplify the lattice-driven effect. However, this measurement starts from a state where hexagonal frames are already present. To further examine the effect of the magnetic field, the CGT crystals were first cooled to 60 K under zero field, followed by a field-cycling measurement (Figure 4; Figure S7 for the complete set of images). An image with a relatively large scan area was collected, as shown in Figure 4a. The whole sample exhibits a stripe-domain ground state, but closed-loop elliptical and polygonal magnetic structures are also observed, as circled by the red dashed lines. These observations highlight the excellent reproducibility of the results, as compared with the ZFC data shown in Figure 2. To reveal more details of the spin structures, the scan size was reduced for the subsequent images. As the magnetic field increases (Figures 4b–e), the stripe domains magnetized along the field direction grow and ultimately dominate the entire sample area. Starting from this “pure” ferromagnetic state, we lower the field and observe the sudden emergence of a hexagonal ring structure (Figure 4f). As the field decreases, the originally straight edges of the hexagonal ring structure begin to bend (Figures 4g,h), and at zero field, the entire ring becomes fully distorted. However, it still retains a closed-loop configuration, thereby preserving its topological nature.

Subsequently, a reversed field was applied, as illustrated in Figures 4j–o. Interestingly, the distorted ring structure gradually recovered its hexagonal form with increasing field and was fully reconstructed at −1000 Oe. This further confirms that the formation of the hexagonal structure is indeed driven by magnetic anisotropy. Dramatic changes occurred at −1200 Oe, where the HST irreversibly broke up into stripe domains. Subsequently, these stripe domains broke into isolated circular magnetic structures and eventually reached a saturated magnetic state. Note that the size of the hexagonal rings driven by the magnetic field is much larger than those induced by temperature. As shown in Figures 2 and 3 (temperature-driven), the edge length of the hexagons is approximately 800 nm, whereas it reaches up to 2300 nm in Figure 4 (field-driven). It can be inferred that the vdW lattice dominates the spin ordering process, while the magnetic field serves as an auxiliary factor that acts upon the preexisting lattice structure.

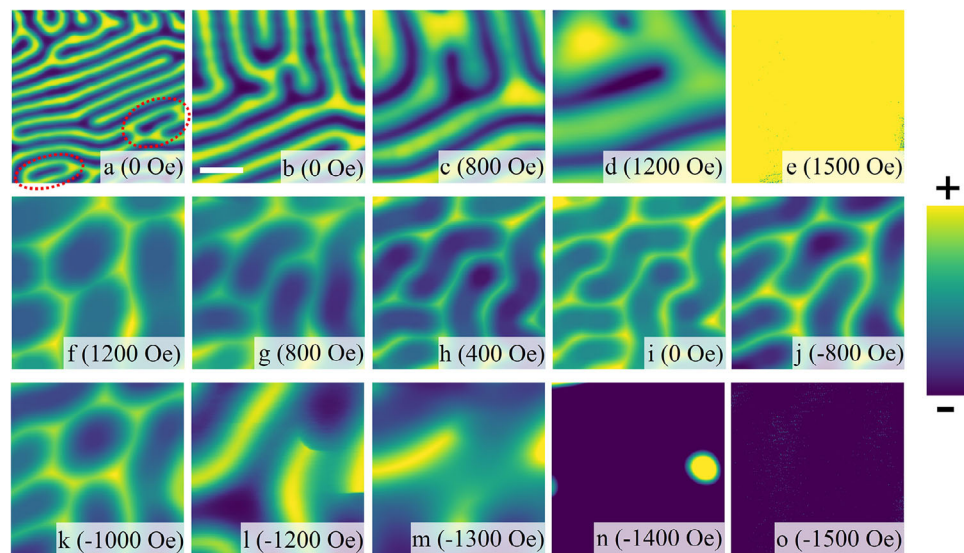


FIGURE 4 | Field-cycling images reveal the dynamic evolution of field-induced HST. (a) MFM image taken as a relatively scan scale of 11 μm . (b–o) A continuous set of MFM images taken at the same location with a scan scale of 8 μm . The scale bar in Figure 4b is 2 μm . The average area of the hexagonal rings under different magnetic fields is 14 μm^2 , and the average edge length is 2.3 μm . The sample was first cooled from 120 K to 60 K in zero magnetic field, after which the field was applied.

5 | Dynamic Behavior of Topological Structures

While topologically nontrivial spin textures such as skyrmions in CGT have been extensively studied through high-resolution imaging techniques such as Lorentz TEM, lattice-correlated topological spin textures have, to the best of our knowledge, not been reported previously [17, 23, 63]. In the absence of antisymmetric exchange interactions such as the Dzyaloshinskii-Moriya interaction (DMI), which are prohibited in centrosymmetric CGT, the formation of the observed skyrmion lattice as shown in Figure 1d is ascribed to the interplay between uniaxial magnetocrystalline anisotropy and magnetic dipolar interactions. More specifically, the observed skyrmions can be attributed to type-I magnetic bubbles, which are characterized by continuous Bloch walls without the presence of Bloch lines or points. They have a cylindrical shape and can be significantly modulated by external magnetic fields. Such skyrmionic bubbles, however, have also been observed by our MFM measurements on another CGT crystal (Figure S3). Although the measurements were performed under the same experimental conditions, distinct magnetic structures were observed, suggesting that the emergence of HSTs depends on delicate lattice interactions, which are highly sensitive to the sample's condition, such as its crystallinity (Figures S8, S9 and S10) [48].

In the following, we examine the topological characteristics of the morphologically diverse polygonal structures observed in CGT. First, we have to admit that the topological nature of the HSTs cannot be unambiguously determined solely from MFM measurements. This question could be specifically addressed by studies using magnetic imaging techniques with higher spatial resolution, such as Lorentz transmission electron microscopy (LTEM). It has been reported using LTEM that a hexagonal lattice of skyrmion bubbles can be obtained by field cooling [23]. Upon an optimized field cooling procedure, hexagonal skyrmion

lattices are also successfully generated in other 2D nanoflake materials like Fe_3GeTe_2 [20, 50, 51]. We have also observed a similar hexagonal skyrmion lattice in another CGT sample (Figure S3). Therefore, we can conclude that the HSTs we observed are located in the same region as the hexagonal skyrmion lattice in the phase diagram, and should also possess a topological nature. In other words, we consider them to be different morphological forms with the same topological nature, and they are topologically equivalent. The non-hexagonal skyrmion polygons are ascribed as lattice defects in a skyrmion crystal, appearing as edge dislocations [64].

The clear and well-defined boundaries make them an excellent model system for exploring topological phenomena. As demonstrated in Figures 4f–k, once established, the topological spin texture maintains its stability under external perturbations through adaptive distortions or smooth shape transformations. It exhibits topological stability across a broad magnetic field range (up to 1200 Oe), close to the saturation field of 1500 Oe. More intriguingly, the hexagonal magnetic structure preserves nearly the same size during the entire field cycling process, as evidenced by Table S2 in the Supporting Information. We performed statistical analysis of the hexagonal areas and found that they remain nearly constant at approximately 14 μm^2 across different magnetic fields. This behavior contrasts sharply with that of the trivial stripe domains (Figures 4b–d), in which the domain width expands as the magnetic field increases. The robust topological nature is further supported by the temperature-dependent behavior as shown in Figures 2e–h: once nucleated, the topological spin textures undergo distortions in response to temperature variations, yet their topological properties are preserved. The topological spin textures exhibit remarkable stability over a broad range of parameter space, from 69.5 K down to 5 K and near the saturation field (noting that the T_C is only 66 K; see Figure 1c).

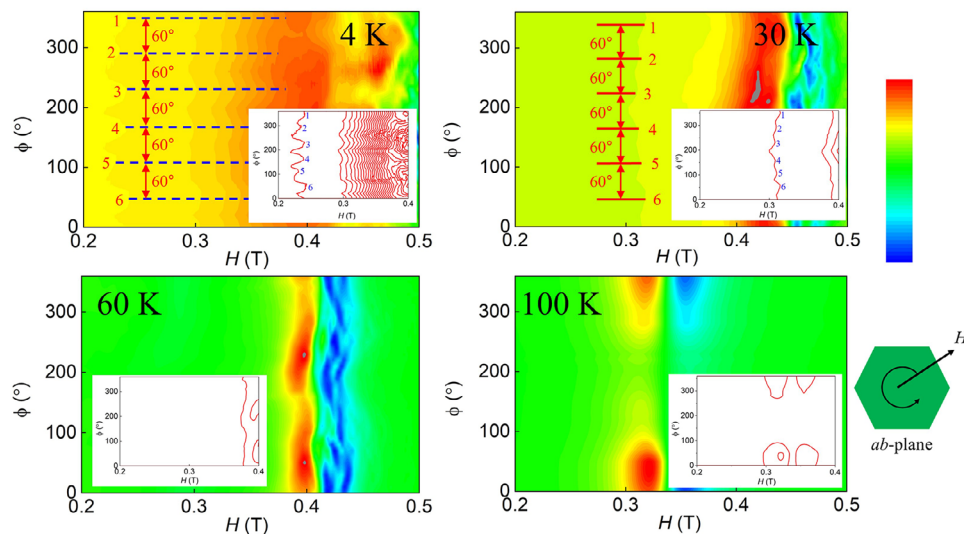


FIGURE 5 | ESR spectra of the same CGT crystal at representative temperatures. Magnetic field applied in the *ab*-plane and measured over 360° with 10° intervals. The initial orientation of the magnetic fields does not align with a specific axis of the sample. Insets of each figure present the contour plots of the ESR spectra.

The collapse dynamics of topological spin textures, although extensively studied theoretically, have rarely been observed experimentally [65–69]. The relatively large size and well-defined boundaries provide an excellent platform for studying the dynamic processes of topological spin textures. Here, the core annihilation process of a single topological texture is fully captured by our MFM imaging, revealing critical details of its instability and collapse mechanisms that were previously inaccessible. As shown in Figure 4l, the topological texture collapses into stripe domains after an additional 200 Oe field is applied. Note that the stripe domains, which serve as the boundaries of the topological textures, have largely shifted in position. In sharp contrast, during the subsequent melting process shown in Figures 4m–o, the trivial stripe domains first fragment into short strips, then transform into skyrmionic bubbles, and finally reach a saturated state, all without significant positional changes. This suggests that the topological texture undergoes collapse in a manner resembling the behavior of a quasiparticle, a concept that has been intensely studied in current-driven dynamics. The magnetic moments are compressed by the competition of magnetic interactions to form topological spin structures. Upon collapse, the release of this compression leads to the displacement of the boundary moments, driven by the spin-transfer torque [66]. This behavior highlights the inherently robust and invariant nature of topological spin textures, which enables them to exhibit quantized responses to continuous perturbations.

6 | Physical Formation Mechanisms

To gain deeper insight into the correlation between magnetic spins and the lattice symmetry/structural environment, we performed temperature-dependent ESR measurements, as presented in Figure 5 [55]. Since CGT is ferromagnetic, the ESR measurements correspond to FMR, which reflects the collective magnetization dynamics of the CGT crystal [44, 45]. The magnetic field was oriented in the *ab*-plane, and measurements were taken over a full 360° rotation, as depicted in the schematic of Figure 5.

ESR spectra of CGT were measured at 4, 30, 60, and 100 K. At 4 K, the spectrum exhibits pronounced six-fold peaks, reflecting the strong coupling between ferromagnetic multidomain collective excitations and the underlying lattice symmetry. This observation is consistent with our MFM imaging, where hexagonal magnetic domain patterns are stabilized at low temperature, demonstrating a real-space manifestation of the lattice-imposed spin symmetry. At 30 K, the six-fold periodicity remains discernible but the peak intensity is significantly reduced, indicating the weakening of magnetic anisotropy and domain coherence. Near the Curie temperature (~60 K), the six-fold structure completely vanishes and only a continuous resonance line persists, characteristic of the collapse of the HSTs. At 100 K, well above the Curie point, the resonance line further fragments, evidencing the fully paramagnetic state where local spin fluctuations dominate. These results highlight that the six-fold ESR modulation directly encodes the interplay between spin collective modes and the crystalline lattice symmetry, and its thermal evolution provides microscopic insight into the melting of HSTs across the phase transition.

It is widely reported that in the 2D regime, magnetic anisotropy can open a spin-wave excitation gap, which is crucial for stabilizing magnetic order. Although our MFM measurements are conducted on bulk single crystals, the intrinsic layered nature of CGT gives rise to quasi-2D magnetic behavior. In such systems, long-range magnetic order is still vulnerable to thermal fluctuations. However, the strength of the uniaxial anisotropy effectively lifts the degeneracy of low-energy spin excitations, stabilizing long-range magnetic order and facilitating the emergence of well-defined spin textures. In particular, the suppression of irregular spin fluctuations enables the formation of intrinsic magnetic textures that are strongly constrained by the symmetry of the 2D ferromagnetic honeycomb lattice. This is evidenced by the observation of HSTs in our MFM images, which reflect the underlying crystallographic symmetry of CGT. These structures likely represent a spontaneous self-organization of spins stabilized by the combined effect of anisotropy and lattice constraints.

In conclusion, lattice-driven topological spin textures with a hexagonal ring structure have been observed in a vdW ferromagnet. These exotic spin textures, characterized by their regular shape, are induced by the enhancement of intrinsic uniaxial magnetocrystalline anisotropy and are strongly correlated with the symmetry of the hexagonal ring network at the atomic level. The topological stability, rich morphological diversity, and tunable lattice coupling of these spin textures offer significant potential for future developments in spintronics and related device applications.

7 | Methods

7.1 | Single-Crystal Growth and Characterization

Single-crystalline CGT samples were synthesized via the self-flux method, with additional information on their fundamental physical properties available in Ref. [55]. Magnetic properties were characterized using a Quantum Design Magnetic Property Measurement System (MPMS3). ESR measurements were carried out using a Bruker EMXplus 10/12 spectrometer equipped with an Oxford ESR910 liquid-helium cryostat.

7.2 | MFM Experiment and Data Processing

MFM measurements were performed using a custom-designed system capable of variable-temperature operation and high magnetic fields, equipped with an 18 T superconducting magnet (Oxford Instruments). A home-built cryogen-free cryostat was used for low-temperature experiments (Temp Nanjing industrial technology Co.Ltd). A commercial piezoresistive cantilever (PRSA-L300-F50-STD, SCL-Sensor Tech) served as the force sensor. The tip was sequentially coated with 5 nm of titanium, 50 nm of cobalt, and 5 nm of gold by electron-beam evaporation. Prior to installation on the scan head, the tip was magnetized perpendicular to the cantilever plane using a permanent magnet. The cleavage and transfer of the CGT bulk sample to the MFM scan head were both carried out in a glovebox, effectively avoiding exposure to air and preventing degradation of the pristine surface. Data analysis was conducted using WSxM and Gwyddion software.

7.3 | Micromagnetic Simulations

Micromagnetic simulations were performed using the GPU-accelerated MuMax3 package to investigate the formation and stability of spin textures in CGT. The material parameters were chosen according to experimental reports and previous theoretical estimations.

The parameters used for the simulation of the skyrmion bubble lattice are listed below. The sample thickness was set to 100 nm to represent the bulk-like characteristics of the material. The simulated lateral dimensions were set between 1 μm and 5 μm to balance computational cost and simulation accuracy, with a uniform cell size of 5 nm $\times$ 5 nm $\times$ 5 nm. Periodic boundary conditions were applied in the in-plane directions to minimize finite-size effects,

and an edge smoothing factor of 8 was used to suppress boundary artifacts.

The saturation magnetization was set to $M_s = 3 \times 10^5$ A/m. The exchange stiffness A_{ex} was swept from 1×10^{-12} to 6×10^{-12} J/m, and the uniaxial anisotropy constant K_{ul} ranged from 0.1 to 5×10^4 J/m³, with the easy axis oriented perpendicular to the film plane. The Gilbert damping constant was fixed at $\alpha = 0.02$. To mimic the field-cooling process, the initial magnetization configuration was randomized, corresponding to a high-temperature paramagnetic state. Subsequently, an external magnetic field was applied along the out-of-plane direction, and the system was relaxed to its equilibrium state using the relax command. The resulting magnetization configurations after relaxation were used to identify the stabilized spin textures and analyze their dependence on anisotropy and exchange stiffness. In the subsequent MFM simulations, the external magnetic field was set to match the experimental measurement conditions, for instance 100 Oe applied along the out-of-plane direction.

For the simulation of HSTs, hexagonal boundary confinements were applied. The side length and thickness of the hexagon are 600 nm and 50 nm, respectively.

Author contributions

Q.Y.F. and S.D. performed the MFM measurements and characterized the materials, including the transport and magnetic measurements; X.L. and Y.P.S. synthesized the CGT single crystals; Q.Y.F. and M.Y.M. carried out the micromagnetic simulations; Q.Y.F., C.H.X., and Q.Y.L. performed the data analysis and wrote the paper; All authors participated in the discussion of the results. Q.Y.F. and Q.Y.L. supervised the project.

Acknowledgements

This work was financially supported by the National Key R&D Program of China (Grants 2023YFA1607701 and 2023YFA1607702), the National Key Research and Development Program under Contracts of China (Grants 2021YFA1600201 and 2023YFA1607402), the HFNL Self-Deployed Project (Grant ZB2025020100), the National Natural Science Foundation of China (Grants 12527803, 51627901, and 12274412), the Chinese Academy of Sciences Scientific Research Equipment (Grant YZ201628), the Systematic Fundamental Research Program Leveraging Major Scientific and Technological Infrastructure, Chinese Academy of Sciences under Contract (Grant JZHKYPT-437 2021-08), and the Maintenance and Renovation Project for CAS Major Scientific and Technological Infrastructure (Grant DSS-WXGZ-2021-0006). We thank the Hefei Advanced Computing Center for its support. We thank the staff members of the Electron Spin Resonance System (<https://cstr.cn/31125.02.SHMFF.ESR>) at the Steady High Magnetic Field Facility, CAS (<https://cstr.cn/31125.02.SHMFF>), for providing technical support and assistance in data collection and analysis.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. K. S. Novoselov, A. K. Geim, S. V. Morozov, et al., "Electric Field Effect in Atomically Thin Carbon Films," *Science* 306 (2004): 666–669.
2. K. S. Novoselov, A. K. Geim, S. V. Morozov, et al., "Two-dimensional Gas of Massless Dirac Fermions in Graphene," *Nature* 438 (2005): 197–200.
3. Y. Zhang, Y. W. Tan, H. L. Stormer, and P. Kim, "Experimental observation of the quantum Hall effect and Berry's phase in graphene," *Nature* 438 (2005): 201–204.
4. H. Zhou, N. Auerbach, M. Uzan, et al., "Imaging Quantum Oscillations and Millitesla Pseudomagnetic Fields in Graphene," *Nature* 624 (2023): 275–281.
5. T. Han, Z. Lu, Z. Hadjri, et al., "Signatures of Chiral Superconductivity in Rhombohedral Graphene," *Nature* 643 (2025): 654–661.
6. Y. Cao, V. Fatemi, S. Fang, et al., "Unconventional Superconductivity in Magic-Angle Graphene Superlattices," *Nature* 556 (2018): 43–50.
7. Z. Lu, T. Han, Y. Yao, et al., "Fractional Quantum Anomalous Hall Effect in Multilayer Graphene," *Nature* 626 (2024): 759–764.
8. X. Xu, J.-X. Yin, W. Ma, et al., "Topological Charge-entropy Scaling in Kagome Chern Magnet TbMn_6Sn_6 ," *Nature Communications* 13, no. 1 (2022), <https://doi.org/10.1038/s41467-022-28796-6>.
9. C. Gong, L. Li, Z. Li, et al., "Discovery of Intrinsic Ferromagnetism in Two-dimensional van der Waals Crystals," *Nature* 546 (2017): 265–269.
10. B. Huang, G. Clark, E. Navarro-Moratalla, et al., "Layer-dependent Ferromagnetism in a van der Waals Crystal down to the Monolayer Limit," *Nature* 546 (2017): 270–273.
11. N. D. Mermin and H. Wagner, "Absence of Ferromagnetism or Antiferromagnetism in One- or Two-dimensional Isotropic Heisenberg Models," *Physical Review Letters* 17 (1966): 1133–1136.
12. A. Vervelaki, K. Bagani, D. Jetter, et al., "Visualizing Thickness-dependent Magnetic Textures in Few-layer $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Communications Materials* 5 (2024): 40.
13. T. Song, Q. C. Sun, E. Anderson, et al., "Direct Visualization of Magnetic Domains and Moiré Magnetism in Twisted 2D Magnets," *Science* 374 (2021): 1140–1144.
14. Y. Chen, K. Samanta, N. A. Shahed, et al., "Twist-assisted All-antiferromagnetic Tunnel Junction in the Atomic Limit," *Nature* 632 (2024): 1045–1051.
15. Y. Xu, A. Ray, Y. T. Shao, et al., "Coexisting ferromagnetic-antiferromagnetic state in twisted bilayer CrI_3 ," *Nature Nanotechnology* 17 (2022): 143–1479.
16. F. Yao, V. Multian, Z. Wang, et al., "Multiple Antiferromagnetic Phases and Magnetic Anisotropy in Exfoliated CrBr_3 Multilayers," *Nature Communications* 14 (2023): 4969.
17. M. Khela, M. Dabrowski, S. Khan, et al., "Laser-induced Topological Spin Switching in a 2D van der Waals Magnet," *Nature Communications* 14 (2023): 1378.
18. W. Zhu, C. Song, L. Han, T. Guo, H. Bai, and F. Pan, "Van der Waals Lattice-induced Colossal Magnetoresistance in $\text{Cr}_2\text{Ge}_2\text{Te}_6$ Thin Flakes," *Nature Communications* 13 (2022): 6428.
19. Q. Tong, et al. *Skyrmions in the Moiré of van der Waals 2D Magnets* 18 (2018): 7194–7199.
20. H. Zhang, D. Raftrey, Y. T. Chan, et al., "Room-temperature skyrmion lattice in a layered magnet ($\text{Fe}_{0.5}\text{Co}_{0.5}$) $_2\text{GeTe}_2$," *Science Advances* 8 (2022): abm7103.
21. F. Hendriks, R. R. Rojas-Lopez, B. Koopmans, and M. H. D. Guimarães, "Electric Control of Optically-induced Magnetization Dynamics in a van der Waals Ferromagnetic Semiconductor," *Nature Communications* 15 (2024): 1298.
22. M. Dabrowski, S. Haldar, S. Khan, et al., "Ultrafast Thermo-optical Control of Spins in a 2D van der Waals Semiconductor," *Nature Communications* 16 (2025): 2797.
23. M. G. Han, J. A. Garlow, Y. Liu, et al., "Topological Magnetic-spin Textures in Two-dimensional van der Waals $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Nano Letters* 19 (2019): 7859–7865.
24. J. D. Thomsen, M. G. Han, A. N. Penn, et al., "Effect of Surface Oxidation and Crystal Thickness on the Magnetic Properties and Magnetic Domain Structures of $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *ACS Nano* 18 (2024): 13458–13467.
25. A. O'Neill, S. Rahman, Z. Zhang, et al., "Enhanced Room Temperature Ferromagnetism in Highly Strained 2D Semiconductor $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *ACS Nano* 17 (2023): 735–742.
26. Z. Wang, T. Zhang, M. Ding, et al., "Electric-field Control of Magnetism in a Few-layered van der Waals Ferromagnetic Semiconductor," *Nature Nanotechnology* 13 (2018): 554–559.
27. Y. Yin, M. Liu, W. Li, et al., "Manipulation of Room-temperature Magnetic Skyrmions in a van der Waals Ferromagnet Fe_3GaTe_2 ," *New Journal of Physics* 27 (2025): 013023.
28. H. Shi, J. Zhang, Y. Xi, et al., "Dynamic Behavior of above-room-temperature Robust Skyrmions in 2D van der Waals Magnet," *Nano Letters* 24 (2024): 11246–11254.
29. Z. Li, H. Zhang, G. Li, et al., "Room-temperature sub-100 nm Néel-type skyrmions in non-stoichiometric van der Waals ferromagnet $\text{Fe}_3\text{xGaTe}_2$ with ultrafast laser writability," *Nature Communications* 15 (2024): 1017.
30. X. Z. Yu, W. Koshibae, Y. Tokunaga, et al., "Transformation between Meron and Skyrmion Topological Spin Textures in a Chiral Magnet," *Nature* 564 (2018): 95–98.
31. D. Y. Wang, Q. Jiang, W. J. Liu, et al., "In-plane Momentum-dependent Spin Texture Originating from Surface Symmetry Breaking in the Two-dimensional van der Waals Ferromagnet CrGeTe_3 ," *Physical Review B* 107 (2023): 125148.
32. H. Idzuchi, A. E. Llacsahuanga Allica, A. K. A. Lu, et al., "Enhanced Ferromagnetism in an Artificially Stretched Lattice in Quasi-two-dimensional $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Physical Review B* 111 (2025): L020402.
33. S. Samanta, H. Iturriaga, T. T. Mai, et al., "Spin-Phonon Coupling and Magnetic Transition in an Organic Molecule Intercalated $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Nano Letters* 24 (2024): 9169–9177.
34. A. Noah, H. Alpern, S. Singh, et al., "Interior and Edge Magnetization in Thin Exfoliated CrGeTe_3 Films," *Nano Letters* 22 (2022): 3165–3172.
35. L. Chen, C. Mao, J. H. Chung, et al., "Anisotropic Magnon Damping by Zero-temperature Quantum Fluctuations in Ferromagnetic CrGeTe_3 ," *Nature Communications* 13 (2022): 4037.
36. I. H. Kao, J. Tang, G. C. Ortiz, et al., "Unconventional Unidirectional Magnetoresistance in Heterostructures of a Topological Semimetal and a Ferromagnet," *Nature Materials* 24 (2025): 1049–1057.
37. V. Carreaux, et al., "Crystallographic," *Journal of Physics-Condensed Matter* 7 (1995): 69.
38. Y. Hou, Y. Wei, D. Yang, K. Wang, K. Ren, and G. Zhang, "Enhancing the Curie Temperature in $\text{Cr}_2\text{Ge}_2\text{Te}_6$ via Charge Doping: A First-principles Study," *Molecules (Basel, Switzerland)* 28 (2023): 3893.
39. Z. Hao, H. Li, S. Zhang, et al., "Atomic Scale Electronic Structure of the Ferromagnetic Semiconductor $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Science Bulletin* 63 (2018): 825–830.
40. F. Zhu, L. Zhang, X. Wang, et al., "Topological Magnon Insulators in Two-dimensional van der Waals Ferromagnets CrSiTe_3 and CrGeTe_3 : Toward Intrinsic Gap-tunability," *Science Advances* 7 (2021): abi7532.
41. D. Bhoi, J. Gouchi, N. Hiraoka, et al., "Nearly Room-temperature Ferromagnetism in a Pressure-induced Correlated Metallic state of the van der Waals Insulator CrGeTe_3 ," *Physical Review Letters* 127 (2021): 217203.
42. H. Zhao, Y. Ni, B. Hu, et al., "Mechanical Control of Physical Properties in the van der Waals Ferromagnet $\text{Cr}_2\text{Ge}_2\text{Te}_6$ via Application of Electric Current," *Physical Review B* 106 (2022): L041103.

43. S. Spachmann, A. Elghandour, S. Selter, B. Büchner, S. Aswartham, and R. Klingeler, "Strong Effects of Uniaxial Pressure and Short-range Correlations in $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Physical Review Research* 4 (2022): L022040.
44. S. Khan, C. W. Zollitsch, D. M. Arroo, et al., "Spin Dynamics Study in Layered van der Waals Single-crystal $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Physical Review B* 100 (2019): 134437.
45. J. Zeisner, A. Alfonsov, S. Selter, et al., "Magnetic Anisotropy and Spin-polarized Two-dimensional Electron Gas in the van der Waals Ferromagnet $\text{Cr}_2\text{Ge}_2\text{Te}_6$," *Physical Review B* 99 (2019): 165109.
46. I. A. Verzhbitskiy, H. Kurebayashi, H. Cheng, et al., "Controlling the Magnetic Anisotropy in $\text{Cr}_2\text{Ge}_2\text{Te}_6$ by Electrostatic Gating," *Nature Electronics* 3 (2020): 460–465.
47. Z. Q. Wang, F. Xue, L. Qiu, Z. Wang, R. Wu, and Y. Hou, "Switching Intrinsic Magnetic Skyrmions with Controllable Magnetic Anisotropy in van der Waals Multiferroic Heterostructures," *Nano Letters* 24 (2024): 4117–4123.
48. M. T. Birch, F. S. Yasin, K. Litzius, et al., "Influence of Magnetic Sublattice Ordering on Skyrmion Bubble Stability in 2D Magnet Fe_3GeTe_2 ," *ACS Nano* 18 (2024): 18246–18256.
49. M. Wyss, K. Bagani, D. Jetter, et al., "Magnetic, Thermal, and Topographic Imaging with a Nanometer-scale SQUID-on-lever Scanning Probe," *Physical Review Applied* 17 (2022): 034002.
50. B. Ding, Z. Li, G. Xu, et al., "Observation of Magnetic Skyrmion Bubbles in a van der Waals Ferromagnet Fe_3GeTe_2 ," *Nano Letters* 20 (2020): 868–873.
51. C. Liu, S. Zhang, H. Hao, H. Algaidi, Y. Ma, and X. X. Zhang, "Magnetic Skyrmions above Room Temperature in a van der Waals Ferromagnet Fe_3GaTe_2 ," *Advanced Materials* 36 (2024): 2311022.
52. T. E. Park, L. Peng, J. Liang, et al., "Néel-type Skyrmions and Their Current-induced Motion in van der Waals Ferromagnet-based Heterostructures," *Physical Review B* 103 (2021): 104410.
53. T. Guo, Z. Ma, X. Luo, et al., "Multiple Domain Structure and Symmetry Types in Narrow Temperature and Magnetic Field Ranges in Layered $\text{Cr}_2\text{Ge}_2\text{Te}_6$ Crystal Measured by Magnetic Force Microscope," *Materials Characterization* 173 (2021): 110913.
54. X. Liu, Q. Feng, D. Zhang, et al., "Topological Spin Textures in a Non-Collinear Antiferromagnet System," *Advanced Materials* 35 (2023): 2211634.
55. C. Xie, Y. Deng, D. Zhang, et al., "Real-Space Imaging of Intrinsic Symmetry-Breaking Spin Textures in a Kagome Lattice," *Advanced Science* 11 (2024): 2404088.
56. Y. Li, Q. Feng, S. Li, et al., "An Artificial Skyrmion Platform with Robust Tunability in Synthetic Antiferromagnetic Multilayers," *Advanced Functional Materials* 30 (2019): 1907140.
57. Q. Feng, F. Jin, H. Zhou, et al., "Induced Formation of Structural Domain Walls and Their Confinement on Phase Dynamics in Strained Manganite Thin Films," *Advanced Materials* 30 (2018): 1805353.
58. L. Wang, Q. Feng, Y. Kim, et al., "Ferroelectrically Tunable Magnetic Skyrmions in Ultrathin Oxide Heterostructures," *Nature Materials* 17 (2018): 1087–1094.
59. G. T. Lin, H. L. Zhuang, X. Luo, et al., "Tricritical Behavior of the Two-dimensional Intrinsically Ferromagnetic Semiconductor CrGeTe_3 ," *Physical Review B* 95 (2017): 245212.
60. J. Tang, L. Kong, Y. Wu, et al., "Target Bubbles in Fe_3Sn_2 Nanodisks at Zero Magnetic Field," *ACS Nano* 14 (2020): 10986–10992.
61. O. I. Utesov, "Mean-field Description of Skyrmion Lattice in Hexagonal Frustrated Antiferromagnets," *Physical Review B* 105 (2022): 054435.
62. Q. C. Sun, et al., "Magnetic Domains and Domain Wall Pinning in Atomically Thin CrBr_3 Revealed by Nanoscale Imaging," *Nature Communications* 12 (2021): 130601.
63. Y. Lee, J. Yun, G. Kim, et al., "The Magnetic States of a van der Waals Ferromagnet CrGeTe_3 Probed by Vector-field Magnetic Force Microscopy," *Applied Physics Letters* 124 (2024): 130601.
64. R. Brearton, L. A. Turnbull, J. A. T. Verezhak, et al., "Deriving the skyrmion Hall Angle from Skyrmion Lattice Dynamics," *Nature Communications* 12 (2021): 2723.
65. A. Bernand-Mantel, C. B. Muratov, and V. V. Slastikov, "A Micromagnetic Theory of Skyrmion Lifetime in Ultrathin Ferromagnetic Films," *Proceedings of the National Academy of Sciences* 119 (2022): 2122237119.
66. A. Derras-Chouk, E. M. Chudnovsky, and D. A. Garanin, "Dynamics of the Collapse of a Ferromagnetic Skyrmion in a Centrosymmetric Lattice," *Physical Review B* 105 (2022): 134432.
67. A. Derras-Chouk, E. M. Chudnovsky, and D. A. Garanin, "Quantum Collapse of a Magnetic Skyrmion," *Physical Review B* 98 (2018): 024423.
68. F. Rendell-Bhatti, R. J. Lamb, J. W. van der Jagt, G. W. Paterson, H. J. M. Swagten, and D. McGrouther, "Spontaneous Creation and Annihilation Dynamics and Strain-limited Stability of Magnetic Skyrmions," *Nature Communications* 11 (2020): 3536.
69. J. Li, A. Tan, K. W. Moon, et al., "Tailoring the Topology of an Artificial Magnetic Skyrmion," *Nature Communications* 5 (2014): 4704.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm74354-sup-0001-SuppMat.docx.